



AN ANTI-ALIASING ALGORITHM FOR DISCRETE WAVELET TRANSFORM

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Causes of aliasing that inheres in the fast algorithm, the so-called Mallat algorithm, of discrete wavelet transform (DWT) are explained in detail. Also, an anti-aliasing algorithm for DWT is presented. Both the canonical and the presented algorithm are applied to analyse a typical simulated signal and a vibration signal measured from a defective ball bearing. It is shown that the presented algorithm is quite effective for avoiding aliasing and for extracting impact feature from a complex background.

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1. INTRODUCTION

In the last 20 years, wavelet transform (WT) has grown at an explosive rate. Wavelets have appealed to scientists and engineers of many different backgrounds. WT has led to exciting applications in signal analysis and numerical analysis, and many other applications are being studied [1].

The theoretical and computational foundation of WT was laid by some researchers, such as Meyer, Daubechies, Mallat, Coifman and Wickerhanster in the late 1980s and early 1990s [2–10]. The WT of signal $f(t)$ is defined as

$$(W_{\psi}f)(b, a) = |a|^{-1/2} \int_{-\infty}^{\infty} f(t) \bar{\psi}\left(\frac{t-b}{a}\right) dt \quad (1)$$

where ψ is named mother wavelet or basic wavelet and $\bar{\psi}$ is the complex conjugate of ψ . Let

$$\psi_{ab}(t) = \frac{1}{|a|^{1/2}} \psi\left(\frac{t-b}{a}\right) \quad (2)$$

then, ψ_{ab} is the wavelet bases. In equations (1) and (2), the parameter a characterises the frequency of $f(t)$ and is named scale, while the parameter b characterizes the position of $f(t)$ in time or space. All WTs possess four important properties [11]: (1) *Multi-resolution*—One can examine the signal at different time windows and frequency bands by controlling parameters a (dilation factor) and b (translation factor). (2) *Localisation*—By controlling parameters a and b , specific features of a signal at any specific time–frequency window can be obtained explicitly. (3) *Zoom-in and zoom-out*—The wider the time window, the narrower the frequency band, and vice versa. Therefore, WTs are capable of

capturing both the short-time high-frequency information and long-time low-frequency information of the signal. (4) *Reconstruction*—A signal can be reconstructed from its WT at any resolution without information loss. Besides these important and fairly useful properties, moreover, an efficient way to implement the discrete wavelet transform (DWT) was developed by Mallat [5] in 1988. The Mallat algorithm, which is in fact a canonical scheme known in the signal processing community as a two-channel sub-band coder, yields a fast WT and makes the WT have a practical importance [12]. Its applications on various science and engineering fields with great success have been reported in a number of literature.

In some applications of DWT, such as machinery fault diagnosis, a measured signal is expected to be decomposed into a group of sub-band signals by means of DWT, and each sub-band signal is expected within known sub-band. Unfortunately, because of the aliasing inhering in the DWT algorithm [13,14], the ratio of signal to noise (RSN) of each sub-band signal reduces.

By studying and applying DWT, we found that its aliasing problem is significant in some cases. In Section 2 of this paper, we explain the causes of aliasing of DWT in detail. And in Section 3 we present an anti-aliasing DWT algorithm. Finally, we conclude our study in Section 4.

2. CAUSES OF ALIASING INHERING IN MALLAT ALGORITHM

2.1. MALLAT ALGORITHM

Why WT has a practical importance is due to the presence of the so-called Mallat algorithm [5]. Let $A_j[\cdot]$ and $D_j[\cdot]$ be a pair of operators. At j th resolution (or level), $A_j[f(t)]$ and $D_j[f(t)]$ represent the approximation and detail of signal $f(t)$, respectively. The decomposition equations of Mallat algorithm are as follows:

$$\begin{aligned} A_0[f(t)] &= f(t) \\ A_j[f(t)] &= \sum_k H(k-2t) * A_{j-1}[f(t)] \\ D_j[f(t)] &= \sum_k G(k-2t) * A_{j-1}[f(t)] \end{aligned} \quad (3)$$

where $t = 1, 2, \dots, N$, $j = 1, 2, \dots, J$, and $J = \log_2 N$. $H(t)$ and $G(t)$ are a pair of quadrature mirror filters for decomposition: $H(t)$ is a low-pass filter and $G(t)$ is a high-pass filter.

The reconstruction equation of Mallat algorithm is

$$A_j[f(t)] = 2 \left\{ \sum_k h(k-2t) * A_{j+1}[f(t)] + \sum_k g(k-2t) * D_{j+1}[f(t)] \right\} \quad (4)$$

where, $t = 1, 2, \dots, N$, $j = J-1, J-2, \dots, 1, 0$ and $J = \log_2 N$. $h(t)$ and $g(t)$ are a pair of quadrature mirror filters for reconstruction: $h(t)$ is a low-pass filter and $g(t)$ is a high-pass filter.

The canonical DWT algorithm based on Mallat algorithm is shown in Fig. 1. This algorithm is developed into many functions in MATLAB wavelet toolbox. With this algorithm, a series of reconstructions from each single sub-band are obtained, respectively. Because the sampling points of each reconstructed sub-band signal are the same and the same as the original signal, the time resolution of each reconstructed sub-band signal is the same as that of the original signal. This property is fairly useful for some applications such as machinery defect detection.

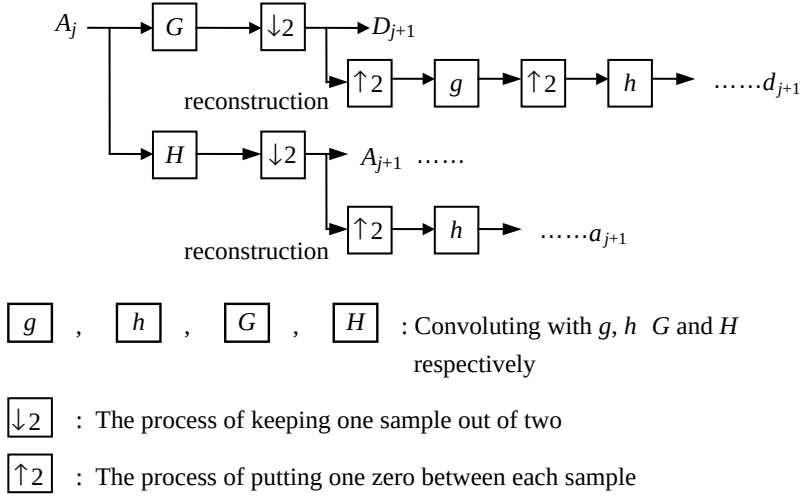


Figure 1. Canonical DWT algorithm.

2.2. ALIASING INHERING IN MALLAT ALGORITHM

From Section 2.1., we can know that downsampling and upsampling are the key steps in the canonical DWT algorithm. But downsampling by itself creates aliases, while upsampling by itself creates images [13]. Therefore, a problem of aliasing inheres in the canonical DWT algorithm. In order to show this problem, let us analyse a typical simulated signal y first by using this algorithm. The signal y is defined as

$$y = \begin{cases} \sin(30\pi t) + \sin(60\pi t) + \sin(90\pi t) + \sin(120\pi t) + \sin(180\pi t) \\ \quad t = [0, 0.125] \cup [0.3725, 0.5] \cup [0.7475, 5.1175] \\ \sin(30\pi t) + \sin(60\pi t) + \sin(90\pi t) + \sin(120\pi t) + \sin(180\pi t) \\ \quad + 2e^{-30t} \sin(260\pi t) \\ \quad t = (0.125, 0.3725) \cup (0.5, 0.7475) \end{cases}$$

Choosing 2048 points from y with a 400 Hz sampling rate and using Daubechies wavelet db4 and calling related functions from MATLAB wavelet toolbox, the signal y is decomposed to level 3. Figure 2 shows the reconstructed sub-band signals and their spectra (for clarity, only partial points of each sub-band signal are shown). For the purpose of comparison, ideal components in each sub-band after applying DWT to signal y are given in Table 1. From Fig. 2 we can find many redundant components in each sub-band. In other words, heavy aliasing appears.

Notice that the related functions used in MATLAB wavelet toolbox are programmed on the basis of the canonical DWT algorithm whose core is the Mallat algorithm. Therefore, it is reasonable to focus our attention on the Mallat algorithm. From equations (3) and (4) and Fig. 1 we can conclude that in the Mallat algorithm there are three key processes: (1) convoluting pre-level WT coefficients with filters of $H(t)$ or $G(t)$ or $h(t)$ or $g(t)$; (2) keeping one sample out of two from a convolution during decomposition; (3) putting one zero between each sample in a pre-level result during reconstruction. Unfortunately, nothing other than the quadrature mirror filters (g, G, h and H), the process of keeping one sample out of two, the process of putting one zero between each sample cause the aliasing.

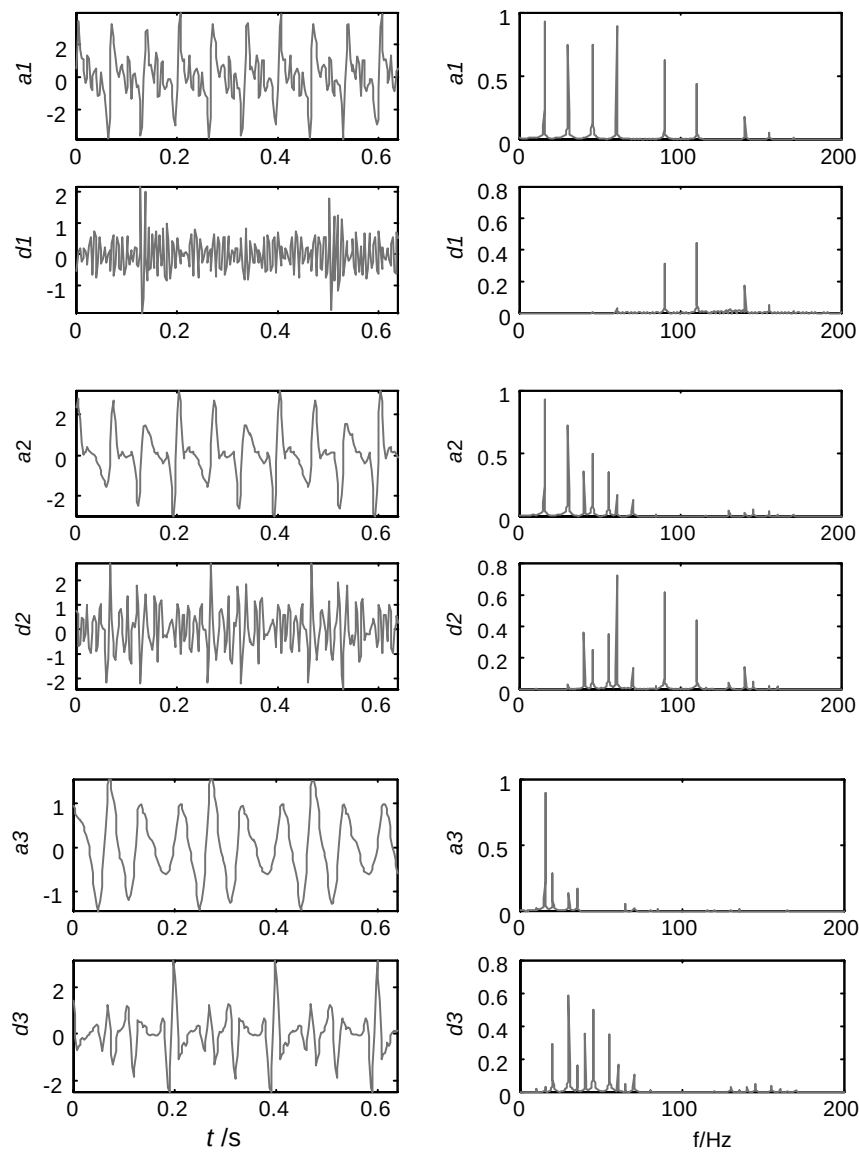


Figure 2. Sub-band signals and their spectra after applying the canonical DWT algorithm with Daubechies wavelets db4 to signal y .

TABLE 1
Ideal components in each sub-band of signal y (Hz)

Levels	1	2	3
a	15, 30, 45, 60, 90	15, 30, 45	15
d	130	60, 90	30, 45

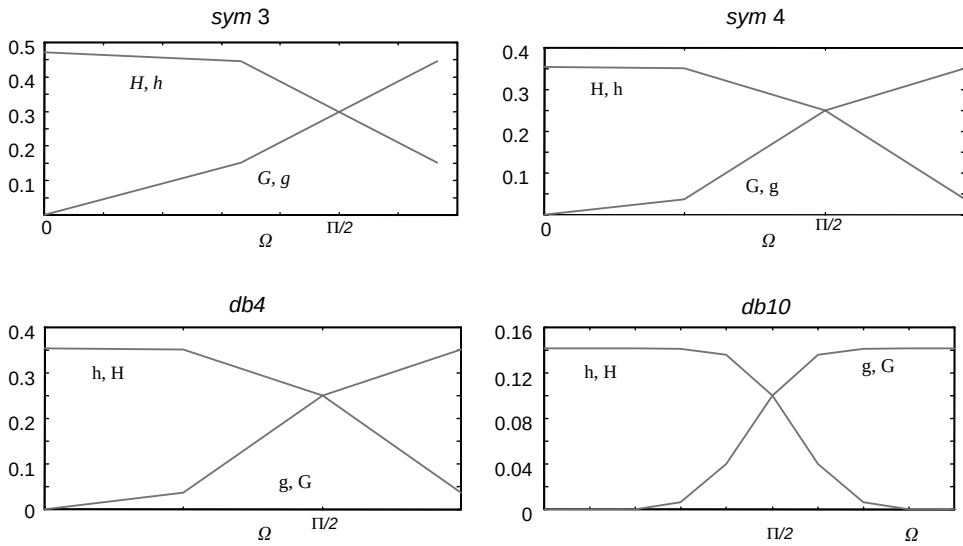


Figure 3. Frequency domain characteristics of quadrature mirror filters of Daubechies wavelets *db4*, *db10* and Symlets wavelets *sym3*, *sym4*.

Quadrature mirror filters. Theoretically, by the Mallat algorithm, a signal is decomposed into a series of sub-bands step by step. In each step, the sub-band width is decomposed fifty-fifty by filter H and filter G . For reconstruction, quadrature mirror filters h or/and g is/are used. Unfortunately the quadrature mirror filters H , h and G , g of any wavelet are not band-limited to $\pi/2$. As an example, the quadrature mirror filters of Daubechies wavelets *db4* and *db10*, Symlets wavelets *sym3* and *sym4* are shown in Fig. 3. This causes two shortcomings: (1) at each level, each sub-band includes some redundant components which should be localised in another sub-band; (2) at each level, when the redundant components included in the low-pass sub-band are downsampled, the so-called sampling theorem is not satisfied. Both the shortcomings are offers of the aliasing.

Keeping one sample out of two. This is a downsampling process whose purpose is to halve the sampling rate. However, shortcomings appear synchronously. Besides the second shortcoming above, at each level, when downsample the high-pass sub-band components apart from the redundant, aliasing occurs because the so-called sampling theorem is dissatisfied. Virtual components appear as mirrors of the actual ones. At 2^j scale (j th level), let f_s be the sampling rate, then $f_s/2^j$ is the symmetrical centre of virtual components and the actual ones.

Putting one zero between each sample. This is an upsampling process whose purpose is to double the sampling rate. In this process, however, just like what happens in the downsampling process, virtual components appear as mirrors of the actual ones. But in this case, the symmetrical centre is $f_s/2^{j+1}$.

3. AN ANTI-ALIASING DWT ALGORITHM

To remove the aliases created by downsampling and to wipe out the images created by upsampling, G Strang and T Nguyen [13] presented an idea to filter the signal: A filter comes before the decimator (downsampling operator) to make the aliasing terms zeros. A filter comes after the expander (upsampling operator) to wipe out the images. Similar to

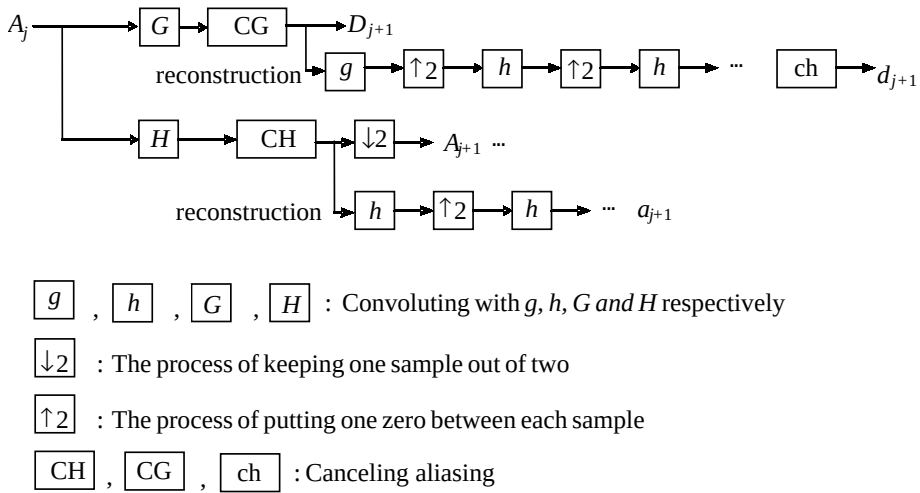


Figure 4. Anti-aliasing DWT algorithm.

this idea, we present an anti-aliasing DWT algorithm. Let j represent the level, A and D represent the approximation and detail, respectively, a represent the reconstruction from one approximation, and d represent the reconstruction from one detail. The anti-aliasing DWT algorithm is shown in Fig. 4. This algorithm is based on the algorithm shown in Fig. 1 and can be regarded as an improved version. With this algorithm, therefore, we can also obtain reconstructions, as same sampling points and same sampling rates as the origin signal, from each sub-band, respectively. But aliasing in each sub-band reconstruction is avoided or removed.

The crucial purpose of the presented anti-aliasing DWT algorithm is to obtain a set of aliasing-free sub-band signals— $a_j, d_j, d_{j-1}, \dots, d_1$ —from the original signal $f(t)$ ($A0$). What we hope the presented anti-aliasing DWT algorithm to fulfill is to increase the RSN of each sub-band signal.

3.1. DOWNSAMPLING AND UPSAMPLING

At the j th level, reconstruct the low-pass sub-band signal before downsampling. Therefore, no upsampling is needed at this level. The process of putting one zero between each sample begins from the next level (2^{j-1} scale). Because no aliasing occurs in the first step of reconstruction, and the virtual components generated in every coming step are filtered by h , the aliasing during calculation of a_j is avoided.

At any level, do not downsample the detail again. Therefore, no upsampling is needed at this level. The process of putting one zero between each sample begins from the next level (2^{j-1} scale). The redundant components created in this sub-band are removed by the operator (ch) at the last step during the reconstruction.

3.2. OPERATOR (CH) AND (CH)

The idea to remove redundant frequency components is as follows: Apply fast Fourier transform (FFT) to related intergrade signals, let the values of redundant frequencies be zeros, then apply inverse fast fourier transform (IFFT) to the resultant spectra. These calculations are expressed by operator (CH), (ch) and (CG) in Fig. 4.

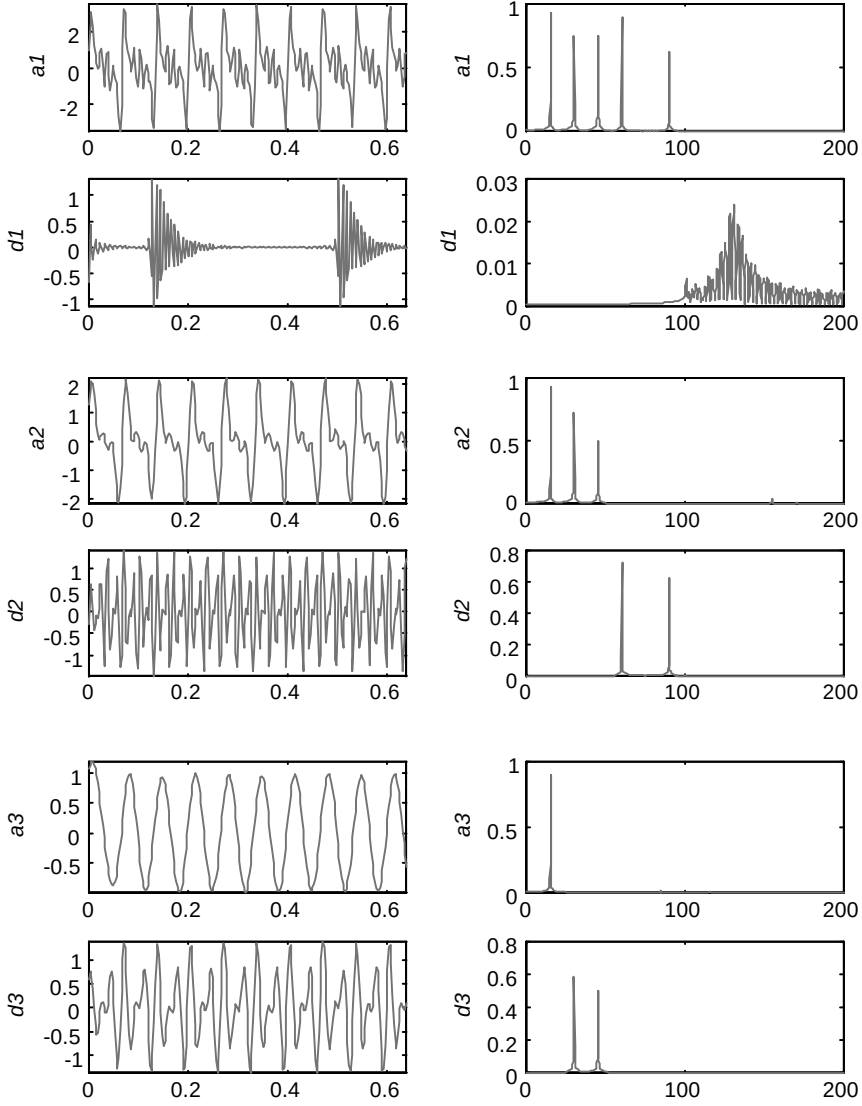


Figure 5. Sub-band signals and their spectra after applying the anti-aliasing DWT algorithm with Daubechies wavelet *db4* to signal *y*.

Let $x(n)$ represent an intergrade signal at the j th level, $W = e^{-j2\pi K/N_j}$, the calculation of (CH) and (ch) is

$$\begin{cases} \bar{X}(k) = \sum_{n=0}^{N_j-1} x(n)W^{kn}, & 0 \leq k \leq \frac{N_j}{4} \quad \text{and} \quad \frac{3N_j}{4} \leq k \leq N_j \\ \bar{X}(k) = 0, & \text{otherwise} \end{cases} \quad (5)$$

$$\bar{x}(n) = \frac{1}{N_j} \sum_{k=0}^{N_j-1} \bar{X}(k)W^{-kn} \quad (6)$$

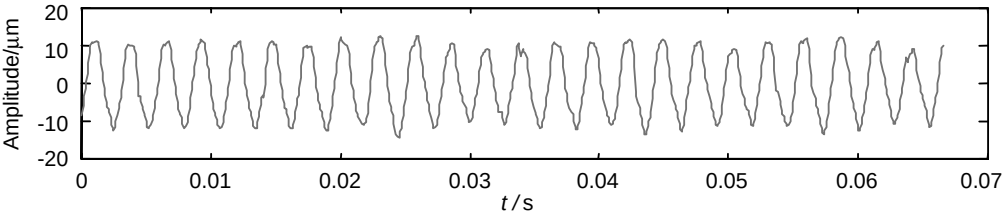


Figure 6. Measured vibration signal x from a defect ball bearing.

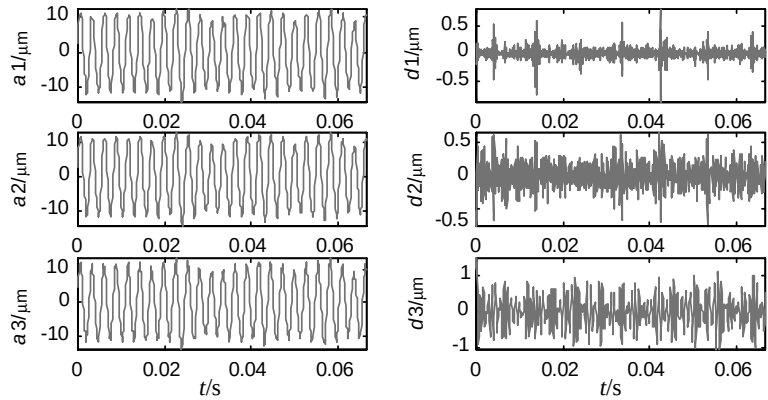


Figure 7. Sub-band signals calculated by applying the anti-aliasing DWT algorithm to a measured vibration signal x .

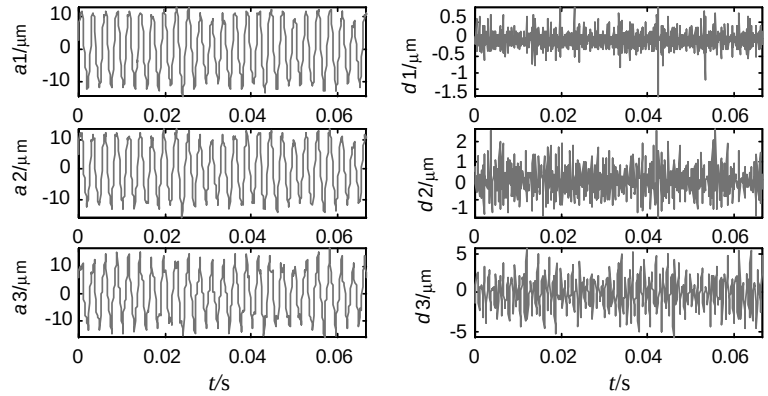


Figure 8. Sub-band signals calculated by applying the canonical DWT algorithm to a measured vibration signal x .

where N_j is the sampling points at the j th level, $k = 0, 1, \dots, N_{j-1}$, $n = 0, 1, \dots, N_{j-1}$, $\bar{x}(n)$ is the output of operator (CH) or (ch). In the case of (ch), N_j is equal to the sampling points of the original signal.

3.3. OPERATOR (CG)

In the case of (CG) in Fig. 4, the calculation is

$$\begin{cases} \bar{X}(k) = \sum_{n=0}^{N_j-1} x(n)W^{kn}, & \frac{N_j}{4} \leq K \leq \frac{3N_j}{4} \\ \bar{X}(k) = 0, & \text{otherwise} \end{cases} \quad (7)$$

where all the symbols are the same as explained in Section 3.2. The inverse transform is calculated according to equation (6).

For implementation, equations (5) and (7) are fulfilled by FFT, and equation (6) by IFFT.

In order to examine the effectiveness of the anti-aliasing DWT algorithm, let us analyse the same signal y defined in 2.2 once again. This time the only difference is that the anti-aliasing DWT algorithm is used. All other conditions are the same as used in Section 2.2. Figure 5 shows the analyzed results. For clarity, a part of each sub-band signal is shown. Contrasting Fig. 5 with Fig. 2, a distinct difference can be seen between them. For instance, two transient components locating at 0.125 and 0.5 s are extracted much more clearly in sub-band $d1$ in Fig. 5 than in Fig. 2. Contrasting every sub-band spectra in the right side of Fig. 5 with Table 1, we can find that the frequency components at each sub-band are exactly the same. The aliasing is avoided rather effectively by the presented anti-aliasing DWT algorithm.

In order to examine the effectiveness of the presented anti-aliasing DWT algorithm for practical uses, we applied it and the canonical DWT algorithm to a measured vibration signal x , respectively. The signal x shown in Fig. 6 was measured from a ball bearing whose inner race has a small dent on its surface. Figure 7 is the DWT result after applying the presented anti-aliasing DWT algorithm to signal x , while Fig. 8 is the DWT result after applying the canonical DWT algorithm to the same signal. In both cases, the wavelet used is Symlets wavelet $sym4$, and the signal is decomposed to level 3. Observing every sub-band signals in Fig. 7, a series of impacts with constant time interval can be seen clearly. Same impacts can also be seen somewhat in $d2$. These impacts typically characterise the type of localised defect in the ball bearing. Observing every sub-band signal in Fig. 8, however, no such impacts can be seen. According to Section 2, the reason is due to the aliasing inhering in the canonical DWT algorithm. Because the anti-aliasing performance of the presented algorithm, the RSN of each resultant sub-band signal is increased, useful information can be extracted into one or several sub-band signals from the raw measured signal.

4. CONCLUSIONS

Three key factors of *quadrature mirror filters, keeping one sample out of two, and putting one zero between each sample* are immanent causes of aliasing in DWT based on the Mallat algorithm. The more complicated an original signal, the heavier the aliasing. Aiming at each aliasing caused factor, the presented anti-aliasing DWT algorithm is designed to suit the remedy to the case. By using the presented DWT algorithm to a complicated simulative signal, it is shown that the aliasing in each sub-band signal is avoided or removed. The critical point for anti-aliasing by the presented DWT algorithm is the frequency resolution. The smaller the frequency resolution, the more effective the anti-aliasing of the presented DWT algorithm. We recommend 1 Hz of frequency resolution for most applications. This means that if a sampling rate is fixed, the same number of sampling points should be chosen and analysed for most applications. In some application

cases, we need to compromise between the effectiveness of anti-aliasing and the calculating duration.

The presented DWT algorithm and the canonical one are also applied to a measured vibration signal, respectively. From this we can conclude that the presented anti-aliasing DWT algorithm is more effective for practical use, such as extracting impact feature from a complex background.

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